EE4-18
EE9-AO6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE PART IV: MEng and ACGI

RADIO FREQUENCY ELECTRONICS

Corrected Copy

Wednesday, 29 April 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : A.S. Holmes

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Special instructions for invigilators

This is a closed book examination.

Special instructions for students

Boltzmann's constant, $k = 1.3805 \times 10^{-23}\ [W.s/K]$

Absolute temperature, $T[K] \approx 273 + T\ [^{\circ}C]$

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The Questions

1. The frequency spectrum from 0.1 to 100 GHz of sky noise temperature is shown in Figure 1.1.

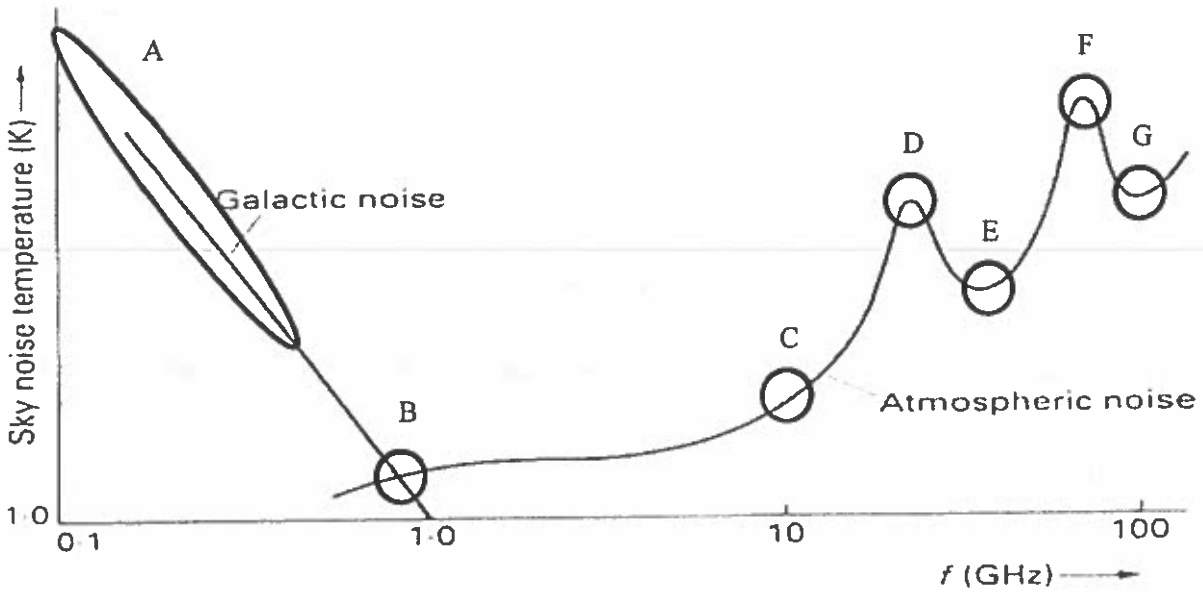


Figure 1.1: Sky noise spectrum.

a) Qualitatively, describe what can be heard when an amplitude modulation radio receiver is not tuned to any broadcasting station and switched from UHF to VHF to HF. Justify possible reasons for what is heard, in terms of the external source and sources within the receiver. For full marks, quote any appropriate equation to support your answer.
[4]

b) State the most common domestic application at Point B on Figure 1.1, indicating its approximate frequency, and briefly explain possible implementation trade-offs at such frequencies. For full marks, quote an appropriate equation to support your answer.
[4]

c) State the most common domestic application at Point C on Figure 1.1. For this application, explain why the uplink frequency is greater than the downlink frequency. For full marks, quote an appropriate equation to support your answer.
[4]

d) Briefly explain the reason for the peak in atmospheric noise at Point D in Figure 1.1 and state the approximate frequency of this peak.
[2]

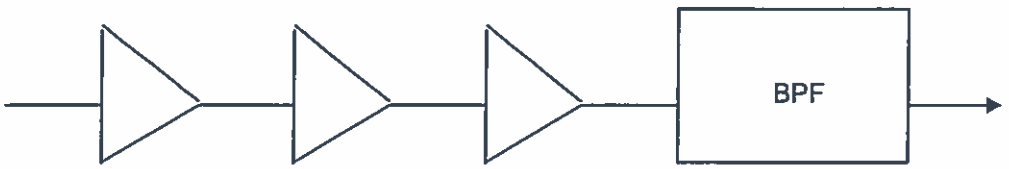
e) State the most common application at Point E on Figure 1.1, indicating its approximate frequency, and briefly explain why this application is in this band.
[2]

f) Briefly explain the reason for the broad peak in atmospheric noise at Point F in Figure 1.1 and state the approximate centre frequency of this broad peak. State the most common domestic application and why it is found in this band.
[2]

g) State the most common application at Point G on Figure 1.1, indicating its approximate frequency, and briefly explain why this application is in this band.
[2]

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2. An amplifier chain is illustrated in Figure 2.1. All sub-systems are assumed to be operating in their linear region and perfectly impedance matched.



$G = 15$ dB	$G = 8$ dB	$G = 4$ dB	$G = -0.7$ dB
$IP_3 = 30$ dBm	$IP_3 = 40$ dBm	$IP_3 = 50$ dBm	
$F = 1$ dB	$F = 2$ dB	$F = 3$ dB	

Figure 2.1: Amplifier chain.

For an overall input power of 3 dBm, calculate the following at the output of each subsystem:

a) C

[2]

b) IP_3

[6]

c) IMD_3

[3]

d) I_3

[3]

e) Overall F

[6]

All variables have their usual meaning.

3.

a) Draw the circuit diagram for a series *RLC* tuned circuit, identifying all the elements, and write down the expression for its total impedance. Assume that the element values are all frequency invariant.

[3]

b) From the total impedance in 3(a), and given the general mathematical identity in equation (3.1), derive an expression for the associated differential-phase group delay for all frequencies.

$$\frac{\partial}{\partial x}\{tan^{-1}(u)\} = \frac{1}{1+u^2}\frac{\partial u}{\partial x} \qquad (3.1)$$

[5]

c) For the element values in 3(a), write down the well-known expressions for:

i) Ideal lossless resonance frequency, ω_l.

[1]

ii) Unloaded quality factor at ω_l.

[1]

d) Using the results in 3(b) and 3(c), prove that the unloaded quality factor at ω_l is directly proportional to the ratio of differential-phase group delay and the time period of the cycle.

[4]

e) Given an nth-stage BPF, constructed using lossless *LC* tuned circuits, the maximum theoretical insertion phase variation is $n\pi$ (as frequency increases from dc to infinity).

i) Derive a qualitative expression for the differential-phase group delay, in terms of the order of the filter and -3 dB bandwidth.

[2]

ii) Using the rough approximation in 3(e)(i), show how the loaded quality factor at centre frequency ω_o is directly proportional to the ratio of differential-phase group delay and the time period of the cycle.

[2]

iii) Comment on the similarity of the expression in 3(e)(ii) for quality factor and that derived in 3(d).

[2]

4.

a) Compare and contrast lumped-element impedance matching over the use of distributed-elements, in terms of frequency performance. How does this affect their role in dc biasing networks?

[4]

b) With lumped-element impedance matching, what terminating impedance conditions are best suited for L-match, π-match and T-match networks?

[4]

c) A 2 GHz narrow-band amplifier has an output impedance of $(5 - j7)\ \Omega$ and must be matched to a system impedance of 50 Ω. Design simple matching circuits to achieve maximum power transfer:

i) With the use of one lumped-element component and one distributed-element component.

[4]

ii) With the use of two lumped-element components.

[4]

d) With MMIC technology, discuss the difficulties implementing lumped-element and distributed-element components. How can micromachining technologies help to overcome some of the problems with monolithic implementations? How does this affect their role in filter networks?

[4]

5.

a) Using S-parameters, what are the levels of insertion loss and return loss at the -3 dB cut-off frequency for a lossless filter? You are asked to design a filter with a maximum pass band return loss level of -6.868 dB. What will be the worst-case pass band insertion loss ripple for a lossless band-pass filter?

[5]

b) Using the worst-case level of ripple calculated in 5(a), but this time for the stop band return loss, design a lumped-element *LC* band stop filter to meet the following specifications:

Lower pass band -3 dB cut-off frequency:	540 MHz
Upper pass band -3 dB cut-off frequency:	660 MHz
Stop band bandwidth:	60 MHz
Band stop attenuation:	> 45 dB
Source impedance, Zs:	50 Ω
Load impedance, Z_l:	100 Ω

[10]

c) Using S-parameters, define differential-phase group delay and explain the general relationship between its frequency response and that of sharp cut-off insertion loss characteristics.

[5]

6. The photograph in Figure 6.1 is of a 38 GHz receiver MMIC.

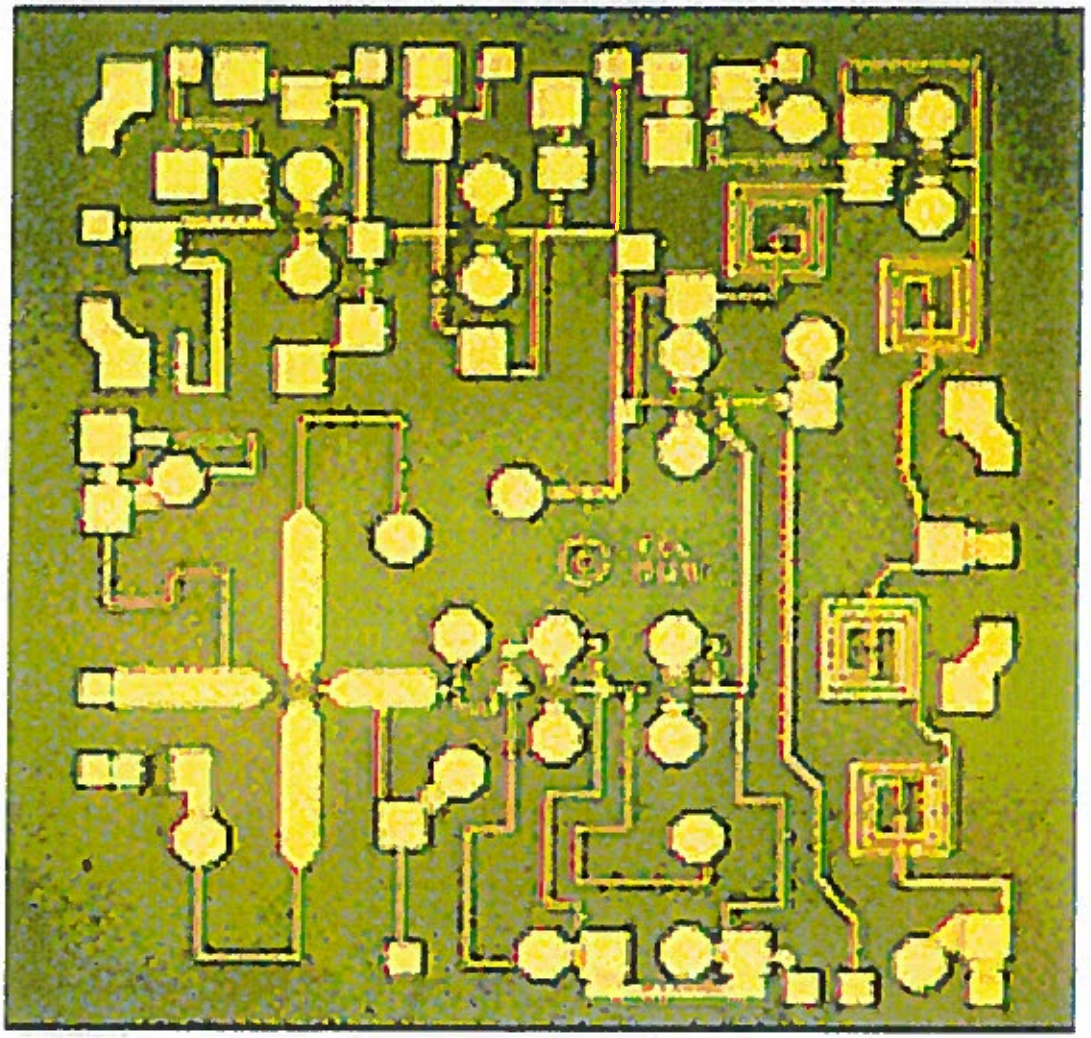


Figure 6.1: Photograph of a 38 GHz receiver MMIC (Nam *et al.*, IEE Coll., 1999).

a) Draw the high level subsystems block diagram from what can be deduced from Figure 6.1. An active transistor should be represented as a discrete amplifier stage.

[4]

b) State the type of receiver that Figure 6.1 represents and list its general characteristics.

[4]

c) What type of mixer is used and how is it configured to operate? Briefly explain what this design is attractive for an MMIC.

[4]

d) If the LO needs the use of a dielectric resonator, where would this resonator be located in practical applications and what is the reasoning for this?

[2]

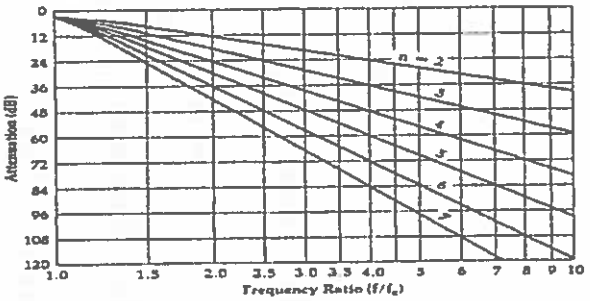
e) In general, what is the rule-of-thumb power level needed for an LO signal at the input of a general mixer used in a receiver and justify the reason for this level?

[2]

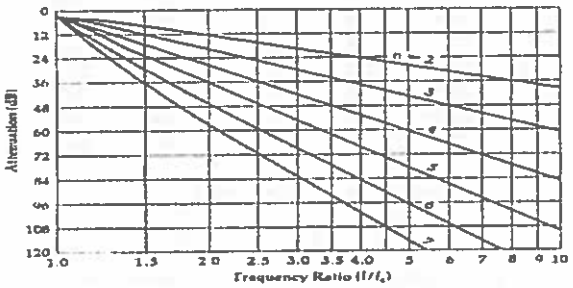
f) For a more complete receiver, what subsystem block is missing from the front end? Explain why this is needed and list three possible solutions to alleviate the problem.

[4]

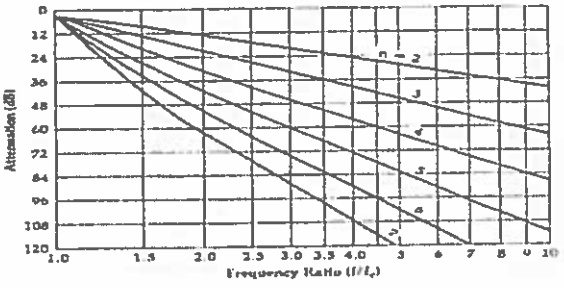
Standard Filter Curves and Tables



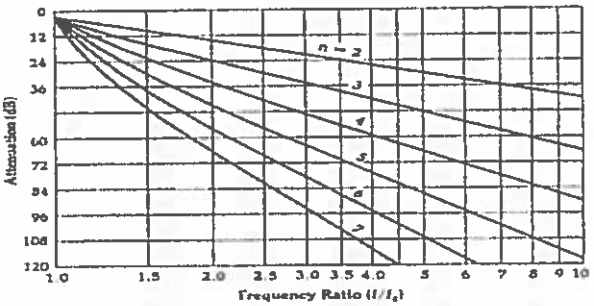
Attenuation characteristics for Butterworth filters.



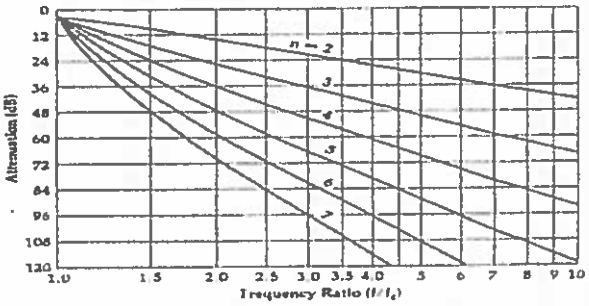
Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.



Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

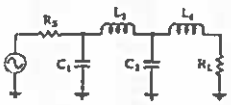


Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.



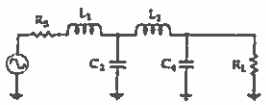
Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

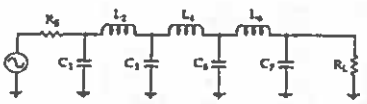


R_s/R_L	C_1	L_2	C_3	L_4
3.000	0.572	3.132		
4.000	0.365	4.600		
8.000	0.157	9.658		
∞	1.213	1.100		
1.000	2.216	1.086	2.216	
0.500	4.431	0.817	2.216	
0.333	5.847	0.726	2.216	
0.250	8.802	0.660	2.216	
0.125	17.725	0.612	2.216	
∞	1.652	1.480	1.106	
3.000	0.853	4.411	0.814	2.535
4.000	0.452	7.083	0.612	2.846
8.000	0.209	17.164	0.428	3.281
∞	1.350	2.010	1.488	1.106

R_L/R_s	L_1	C_2	L_3	C_4

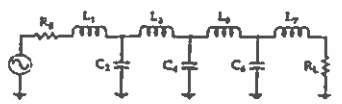


Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple



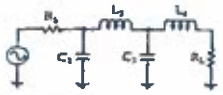
n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	2.207	1.128	3.103	1.128	2.207		
	0.500	4.414	0.565	4.653	1.128	2.207		
	0.333	6.828	0.376	6.205	1.128	2.207		
	0.250	8.829	0.282	7.758	1.128	2.207		
	0.125	17.657	0.141	13.961	1.128	2.207		
	∞	1.721	1.645	2.061	1.493	1.103		
6	3.000	0.870	3.873	0.771	4.711	0.969	2.406	
	4.000	0.481	5.644	0.478	7.351	0.849	2.582	
	8.000	0.227	12.310	0.198	16.740	0.726	2.800	
	∞	1.378	2.097	1.890	2.074	1.494	1.102	
7	1.000	2.204	1.131	3.147	1.194	3.147	1.131	2.204
	0.500	4.408	0.566	6.293	0.895	3.147	1.131	2.204
	0.333	6.813	0.377	9.441	0.706	3.147	1.131	2.204
	0.250	8.815	0.283	12.588	0.747	3.147	1.131	2.204
	0.125	17.631	0.141	25.175	0.671	3.147	1.131	2.204
	∞	1.741	1.677	2.155	1.703	2.079	1.494	1.102

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



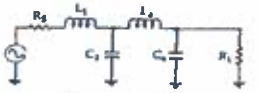
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Butterworth Low-Pass Prototype Element Values

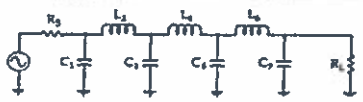


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.005		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.384	1.928	
	0.700	0.915	1.165	2.377	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.501	
	0.400	1.425	0.604	4.664	
	0.300	1.838	0.440	5.563	
	0.200	2.669	0.284	7.910	
	0.100	5.187	0.138	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.468	1.502	1.744	1.459
	1.250	0.388	1.695	1.511	1.811
	1.429	0.325	1.902	1.291	2.175
	1.667	0.269	2.103	1.082	2.613
	2.000	0.218	2.452	0.883	3.187
	2.500	0.160	3.050	0.691	4.009
	3.333	0.124	3.843	0.507	5.338
	5.000	0.080	5.664	0.331	7.940
	10.000	0.039	11.094	0.162	15.642
	∞	1.531	1.577	1.082	0.383

n	R_L/R_s	L_1	C_2	L_3	C_4

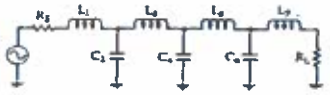


Butterworth Low-Pass Prototype Element Values

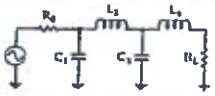


n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	0.900	0.442	1.027	1.910	1.756	1.389		
	0.800	0.470	0.866	2.061	1.544	1.738		
	0.700	0.517	0.731	2.285	1.333	2.108		
	0.600	0.588	0.609	2.600	1.126	2.652		
	0.500	0.696	0.496	3.051	0.924	3.533		
	0.400	0.868	0.388	3.730	0.727	3.965		
	0.300	1.094	0.285	4.884	0.537	5.307		
	0.200	1.608	0.186	7.185	0.352	7.935		
	0.100	3.513	0.091	14.095	0.173	15.710		
	∞	1.545	1.694	1.382	0.894	0.309		
6	1.111	0.289	1.040	1.325	2.054	1.744	1.335	
	1.250	0.245	1.116	1.126	2.230	1.550	1.688	
	1.429	0.207	1.236	0.957	2.499	1.346	2.062	
	1.667	0.173	1.407	0.801	2.858	1.143	2.509	
	2.000	0.141	1.653	0.654	3.369	0.942	3.094	
	2.500	0.111	2.028	0.514	4.141	0.745	3.931	
	3.333	0.082	2.656	0.379	5.433	0.553	5.280	
	5.000	0.054	3.917	0.248	8.020	0.363	7.922	
	10.000	0.028	7.705	0.122	15.786	0.179	15.738	
	∞	1.553	1.759	1.553	1.202	0.758	0.259	
7	0.900	0.290	0.711	1.404	1.489	2.125	1.727	1.290
	0.800	0.322	0.606	1.517	1.278	2.334	1.548	1.652
	0.700	0.357	0.515	1.688	1.091	2.618	1.350	2.028
	0.600	0.408	0.433	1.928	0.917	3.005	1.150	2.477
	0.500	0.480	0.354	2.273	0.751	3.553	0.951	3.084
	0.400	0.590	0.278	2.795	0.592	4.380	0.754	3.904
	0.300	0.775	0.208	3.671	0.437	5.761	0.560	5.258
	0.200	1.145	0.135	5.427	0.287	8.526	0.369	7.908
	0.100	2.257	0.067	10.700	0.142	16.822	0.182	15.748
	∞	1.558	1.799	1.659	1.397	1.055	0.636	0.223

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7

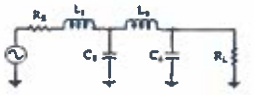


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

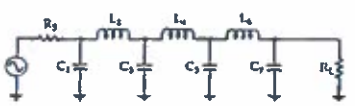


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.101	1.347	1.483		
	1.111	1.347	1.505		
	1.250	0.943	1.937		
	1.429	0.750	2.344		
	1.667	0.600	2.750		
	2.000	0.479	3.277		
	2.500	0.383	4.033		
	3.333	0.259	5.255		
	5.000	0.164	7.650		
	10.000	0.078	14.748		
	∞	1.412	0.742		
3	1.000	1.181	1.821	1.181	
	0.900	1.092	1.660	1.480	
	0.800	1.097	1.443	1.800	
	0.700	1.160	1.228	2.165	
	0.600	1.274	1.024	2.598	
	0.500	1.452	0.829	3.164	
	0.400	1.734	0.645	3.974	
	0.300	2.216	0.470	5.250	
	0.200	3.193	0.305	7.834	
	0.100	6.141	0.148	15.390	
	∞	1.501	1.433	0.591	
4	1.100	0.950	1.936	1.761	1.046
	1.111	0.854	1.840	1.744	1.165
	1.250	0.618	2.075	1.542	1.617
	1.429	0.485	2.278	1.334	2.008
	1.667	0.308	2.571	1.128	2.481
	2.000	0.316	2.994	0.926	3.045
	2.500	0.242	3.641	0.729	3.873
	3.333	0.174	4.727	0.538	5.209
	5.000	0.118	6.910	0.353	7.813
	10.000	0.054	13.469	0.173	15.510
	∞	1.329	1.604	1.212	0.523

n	R_L/R_s	L_1	C_2	L_3	C_4

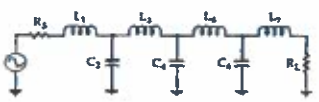


Chebyshev Low-Pass Element Values for 0.01-dB Ripple



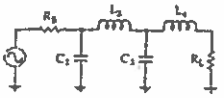
n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	0.977	1.685	2.037	1.685	0.977		
	0.900	0.880	1.456	2.174	1.641	1.374		
	0.800	0.877	1.235	2.379	1.499	1.807		
	0.700	0.926	1.040	2.658	1.323	1.977		
	0.600	1.019	0.863	3.041	1.135	2.424		
	0.500	1.166	0.699	3.584	0.942	3.009		
	0.400	1.395	0.544	4.403	0.749	3.845		
	0.300	1.797	0.398	5.772	0.557	5.193		
	0.200	2.604	0.259	8.514	0.368	7.828		
	0.100	5.041	0.127	16.741	0.182	15.613		
	∞	1.547	1.795	1.645	1.237	0.488		
6	1.101	0.851	1.796	1.841	2.027	1.631	0.937	
	1.111	0.760	1.782	1.775	2.004	1.638	1.053	
	1.250	0.545	1.864	1.489	2.403	1.507	1.504	
	1.429	0.436	2.038	1.266	2.735	1.332	1.899	
	1.667	0.351	2.298	1.061	3.187	1.145	2.357	
	2.000	0.279	2.678	0.867	3.768	0.954	2.948	
	2.500	0.214	3.261	0.682	4.667	0.761	3.790	
	3.333	0.155	4.245	0.503	6.163	0.568	5.143	
	5.000	0.100	6.223	0.330	9.151	0.376	7.785	
	10.000	0.048	12.171	0.162	16.105	0.187	15.595	
	∞	1.551	1.847	1.790	1.598	1.190	0.469	
7	1.000	0.913	1.595	2.002	1.870	2.002	1.595	0.913
	0.900	0.816	1.382	2.089	1.722	2.202	1.581	1.206
	0.800	0.811	1.150	2.282	1.525	2.465	1.464	1.536
	0.700	0.857	0.907	2.516	1.323	2.802	1.307	1.910
	0.600	0.943	0.803	2.872	1.124	3.250	1.131	2.359
	0.500	1.080	0.650	3.382	0.928	3.875	0.947	2.948
	0.400	1.297	0.507	4.156	0.735	4.812	0.758	3.790
	0.300	1.669	0.372	5.454	0.546	6.370	0.568	5.148
	0.200	2.342	0.242	8.057	0.360	9.484	0.378	7.802
	0.100	4.701	0.119	15.872	0.178	16.818	0.186	15.652
	∞	1.559	1.867	1.886	1.765	1.563	1.161	0.456

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



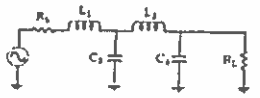
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Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

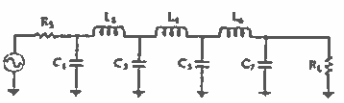


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.355	1.300	1.638		
	1.429	0.977	1.982		
	1.667	0.733	2.489		
	2.000	0.500	3.054		
	2.500	0.417	3.827		
	3.333	0.293	5.050		
	5.000	0.184	7.426		
	10.000	0.097	14.433		
	∞	1.391	0.619		
3	1.000	1.433	1.594	1.433	
	0.900	1.426	1.494	1.622	
	0.800	1.451	1.356	1.871	
	0.700	1.521	1.193	2.190	
	0.600	1.648	1.017	2.603	
	0.500	1.853	0.838	3.159	
	0.400	2.186	0.660	3.968	
	0.300	2.763	0.486	5.279	
	0.200	3.942	0.317	7.850	
	0.100	7.512	0.155	15.466	
	∞	1.513	1.510	0.716	
4	1.355	0.902	2.143	1.585	1.341
	1.429	0.779	2.318	1.429	1.700
	1.667	0.578	2.730	1.165	2.243
	2.000	0.440	3.227	0.907	2.858
	2.500	0.329	3.861	0.700	3.698
	3.333	0.233	5.178	0.500	5.030
	5.000	0.148	7.607	0.367	7.614
	10.000	0.070	14.897	0.180	15.230
	∞	1.511	1.708	1.435	0.673

n	R_L/R_s	L_1	C_2	L_3	C_4

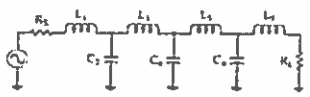


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

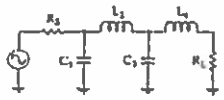


n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.301	1.558	2.241	1.558	1.301		
	0.900	1.285	1.433	2.360	1.486	1.488		
	0.800	1.300	1.282	2.582	1.362	1.736		
	0.700	1.358	1.117	2.868	1.244	2.062		
	0.600	1.470	0.947	3.269	1.085	2.484		
	0.500	1.654	0.778	3.845	0.913	3.055		
	0.400	1.954	0.612	4.720	0.733	3.886		
	0.300	2.477	0.451	6.198	0.550	5.237		
	0.200	3.546	0.295	9.127	0.366	7.489		
	0.100	6.787	0.145	17.937	0.182	13.745		
	∞	1.561	1.607	1.706	1.417	0.651		
6	1.355	0.942	2.080	1.059	2.247	1.534	1.477	
	1.429	0.735	2.349	1.454	2.544	1.405	1.629	
	1.667	0.542	2.800	1.183	3.064	1.185	2.174	
	2.000	0.414	3.368	0.959	3.712	0.979	2.794	
	2.500	0.310	3.765	0.749	4.651	0.778	3.645	
	3.333	0.220	4.927	0.551	6.195	0.580	4.996	
	5.000	0.139	7.550	0.361	9.281	0.384	7.618	
	10.000	0.067	14.220	0.178	18.427	0.190	15.350	
	∞	1.534	1.684	1.831	1.749	1.394	0.638	
7	1.000	1.242	1.520	2.239	1.680	2.239	1.520	1.242
	0.900	1.242	1.395	2.361	1.578	2.397	1.459	1.447
	0.800	1.255	1.245	2.548	1.443	2.624	1.302	1.697
	0.700	1.310	1.083	2.810	1.283	2.942	1.233	2.021
	0.600	1.417	0.917	3.205	1.209	3.384	1.081	2.444
	0.500	1.595	0.753	3.764	0.928	4.015	0.914	3.018
	0.400	1.885	0.593	4.618	0.742	4.970	0.739	3.855
	0.300	2.392	0.437	6.054	0.556	6.569	0.557	5.217
	0.200	3.428	0.286	8.937	0.369	9.770	0.372	7.890
	0.100	6.570	0.141	17.603	0.184	19.370	0.186	15.813
	∞	1.575	1.658	1.921	1.837	1.734	1.379	0.631

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7

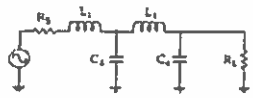


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

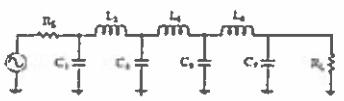


n	R_s/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.050		
	2.000	0.909	2.103		
	2.500	0.504	3.185		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.250	1.834	
	0.900	1.915	1.208	2.028	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.242	
	0.300	3.729	0.463	5.570	
	0.200	5.254	0.309	8.255	
	0.100	9.890	0.153	16.118	
	∞	1.572	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.826
	2.000	0.845	2.720	1.238	1.965
	2.500	0.516	3.706	0.869	3.121
	3.333	0.344	5.120	0.621	4.430
	5.000	0.210	7.706	0.400	6.957
	10.000	0.008	15.352	0.194	14.282
	∞	1.436	1.889	1.521	0.913

n	R_L/R_s	L_1	C_2	L_3	C_4

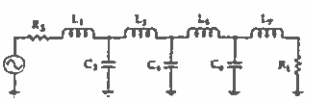


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.691	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.926	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.785	0.942	2.861		
	0.500	2.457	0.754	4.467	0.810	3.414		
	0.400	2.870	0.609	5.598	0.664	4.245		
	0.300	3.589	0.459	8.071	0.506	5.625		
	0.200	5.064	0.306	10.034	0.343	8.367		
	0.100	9.556	0.153	19.647	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.906	2.577	1.386	2.713	1.290	1.706	
	2.000	0.830	2.704	1.291	2.872	1.237	1.856	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.999	0.636	4.481	
	5.000	0.200	7.615	0.406	8.732	0.412	7.031	
	10.000	0.098	15.186	0.197	17.681	0.202	14.433	
7	1.000	1.790	1.296	2.716	1.385	2.716	1.296	1.790
	0.900	1.835	1.215	2.860	1.308	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.213	3.107	1.153	2.168
	0.700	2.011	1.007	3.364	1.105	3.416	1.058	2.455
	0.600	2.174	0.882	3.778	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	4.289	0.814	3.405
	0.400	2.838	0.604	5.293	0.685	5.470	0.689	4.243
	0.300	3.546	0.455	6.807	0.522	7.134	0.513	5.633
	0.200	5.007	0.303	10.049	0.352	10.496	0.348	8.404
	0.100	9.450	0.151	19.849	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_s	L_1	C_2	L_3	C_4	L_5	C_6	L_7



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